



Technical Service Bulletin: TSB220183

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Flywheel Capscrews Breaking Upon Removal

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### Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

### Contents

#### Product Affected

- QSK23 CM500
- QSK23 CM500 K157G

#### Issue Summary

Symptom:

- Flywheel capscrews breaking off after loosening the capscrews 1 revolution.
- The crankshaft threaded holes **not** usable after capscrew shanks removed.

Root Cause:

- Capscrew seizing is caused by capscrew coating layer separation and galling with crankshaft thread.

#### Resolution

If subject engine matches product affected above, reference Service Instructions below for removing flywheel capscrews.

#### Service Instructions

- Prior to removing flywheel capscrews, spray thread penetrant on all capscrews and let the penetrant absorb onto capscrews and crankshaft threads.
- Prior to attempting removal of capscrews, penetrant lube **must** be applied on threads for at least 8 to 12 hours or the durations the penetrant label prescribes.
- For removal process, use an impact wrench to remove the capscrews.
- The combination of penetrant and impact wrench will break down the coating to prevent thread galling.
- See QSK23 Troubleshooting and Repair Manual, Bulletin 4021375. Refer to Procedure 016-005 (/qs3/pubsys2/xml/en/procedures/89/89-016-005.html) in Section 16 for flywheel capscrew clean, inspect, and reuse.

- Prior to installation of capscrews, apply a light film of Molykote™ or equivalent instead of engine oil to the capscrew threads and use new flywheel capscrew with corrosion resistant properties in Table 1.

**Note :** In cases where penetrant soak time of 8 to 12 hours is **not** possible, use an impact wrench to loosen the capscrews.

| Table 1, Service Parts    |                      |          |            |                 |
|---------------------------|----------------------|----------|------------|-----------------|
| Part Description          | Existing Part Number | Obsolete | Superseded | New Part Number |
| Screw, Fracture Resistant | 4096691              | Yes      | Yes        | 5580439         |
| Screw, Fracture Resistant | 4096703              | Yes      | Yes        | 5580435         |

**Part Inventory Action**

Obsolete part numbers in stock are **not** to be used. Obsolete part numbers in stock are to be returned for reimbursement.

**Production Status**

| Table 2, Production Information |                                  |                         |                                  |
|---------------------------------|----------------------------------|-------------------------|----------------------------------|
| Product                         | Engine Serial Number (ESN) First | Build Date <sup>1</sup> | Engine Plant                     |
| QSK23 CM500                     | 85019232                         | 10-May 2024             | HHS - CTIL HIGH-HORSEPOWER (SEZ) |
| QSK23 CM500                     | 85020082                         | 01 Jul 2024             | HHS - CTIL HIGH-HORSEPOWER (SEZ) |

<sup>1</sup>Engine build date can be found on engine dataplate.

**Document History**

| Date      | Details                                            |
|-----------|----------------------------------------------------|
| 2022-8-30 | Module Created                                     |
| 2024-8-21 | Added Table 1, Part Inventory Action, and Table 2. |